Week 3

Keel Bone Chicken Project

01/30/26

PURDUE UNIVERSITY | College of Engineering

This Week

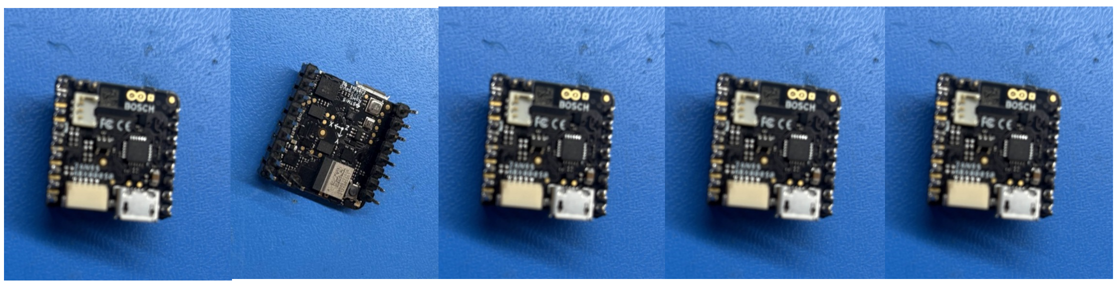
Weekly Progress

- Preparing Devices
  - Soldering Pin Headers onto all Devices
  - Changing Sampling Rates for Devices
- Adjusting Data Collected
  - Changing Time Parameter
  - Adjustable Time Parameter
- PCB Changes
  - PCB Prototyping + Size Changing
  - Parts Ordering

Preparing Devices

Soldering Pin Headers onto all Devices

- After the combined meeting, distribution of devices has been determined:
  - 6 – Agriculture for Testing
  - 3 – Polytechnic for Development
  - 1 – Shorted (Kept for Analysis)
- Currently Soldering <u>5</u> more sets of pin headers onto each device



Preparing Devices

Changing Sampling Rates for Devices

- After discussing with the Agriculture and Mechanical Engineering department, it might be useful to use different sampling rates:
- Sampling rate can be changed in software, selecting:
  - 500 Hz
  - 200 Hz
  - 100 Hz
  - 50 Hz
  - 25 Hz

https://github.com/arduino-libraries/Arduino_BHY2/blob/1.0.3/src/sensors/Se
28

The default parameter values of `SensorClass::begin`:
https://github.com/arduino-libraries/Arduino_BHY2/blob/1.0.3/src/sensors/SensorClass.h#L15 26

```
bool begin(float rate = 1, uint32_t latency = 0);
```

As indicated by the tutorial, `SensorClass::begin` passes those parameters tc `SensorClass::configure`:
https://github.com/arduino-libraries/Arduino_BHY2/blob/1.0.3/src/sensors/SensorClass.cpp#L29 17

```
BHY2.begin(NICLA_STANDALONE);

temperature.begin();
humidity.begin();
pressure.begin();

// High rate sensors configured to 200Hz with 0
gyroscope.begin(200, 0);
accelerometer.beg (int)200
quaternion.begin(200, 0);
```

Adjusting Data Collected

Changing Time Parameter

- Modified Maximum Entries from 1000 → 12000 (For 3 Hours)
- Enough Time for Next Pilot Study
- Will add user input to determine study time when finishing next version of code

```
enable the highlighted flag, and restart Chrome for no
*/

var maxRecords = 12000; // 200 Minutes -> 3 Hours +

var NiclaSenseME =
{
  temperature:
  {
    uuid: '19b10000-2001-537e-4f6c-d104768a1214',
    properties: ['BLERead'],
    structure: ['Float32'],
    data: { temperature: [] }
  },
  humidity:
```

Adjusting Data Collected

Adjustable Time Parameter

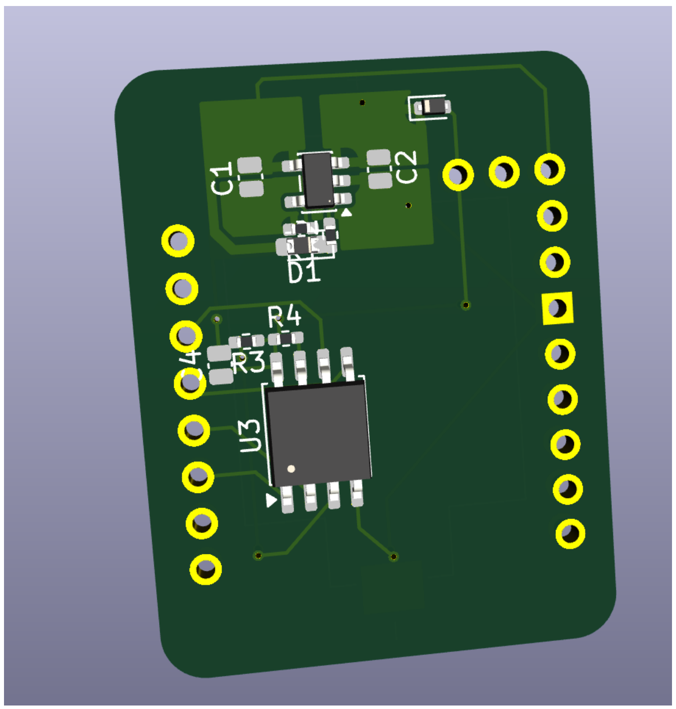
- User input to determine study time when finishing next version of code
- Functionality kept for testing procedures
- Could be replaced in future versions

```python
if __name__ == "__main__":
    # Added Prompt
    while True:
        try:
            minutes_to_save = int(input("Enter number of minutes to save: "))
            if minutes_to_save > 0:
                break
            print("Please enter a positive number.")
        except ValueError:
            print("Please enter a valid number.")

    print(f"Minutes to save: {minutes_to_save}")
    # End Prompt

    server_thread = threading.Thread(target=start_server, daemon=True)
    server_thread.start()

    webview.create_window(
        "Arduino BLE Dashboard",
        f"http://localhost:{PORT}/index.html",
        width=1280,
        height=720
    )
    webview.start()
```

PCB Changes

PCB Prototyping + Size Changing

- Components Resized to fit for Hand Soldering
  - Switched to 04x02 + 05x03 mm^2 Components for Hand Soldering
  - Size on board increased, but very insignificant in general



Symbol : Footprint Assignments

1	BT1 –	LIR2032	: footprints:BAT_504
2	C1 –	4.7 µF	: Capacitor_SMD:C_0504_1310Metric_Pad0.83x1.28mm_HandSolder
3	C2 –	4.7 µF	: Capacitor_SMD:C_0504_1310Metric_Pad0.83x1.28mm_HandSolder
4	C4 –	1 uF	: Capacitor_SMD:C_0504_1310Metric_Pad0.83x1.28mm_HandSolder
5	D1 –	RED	: Diode_SMD:D_0603_1608Metric_Pad1.05x0.95mm_HandSolder
6	D2 –	1N4148WT	: Diode_SMD:D_SOD-523
7	R1 –	2.2 kΩ	: Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder
8	R2 –	470 Ω	: Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder
9	R3 –	10 kΩ	: Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder
10	R4 –	10 kΩ	: Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder
11	U1 –	MCP73831-2-OT	: Package_TO_SOT_SMD:SOT-23-5
12	U2 –	~	: footprints:Arduino_Nicla_Sense
13	U3 –	W25Q128JVS	: Package_SO:SOIC-8_5.3x5.3mm_P1.27mm

PCB Changes

PCB Ordering

- ~13 Components per board x 6 Boards
- Attempting to purchase components from the same vendor (to reduce lead time)

bom

nce	Qty	Value	DNP	Exclude from BOM	Exclude from Board	Footprint	Datashee
	1	LIR2032				footprints:BAT_504	
	2	4.7 µF				Capacitor_SMD:C_0504_1310Metric_Pad0.83x1.28mm_HandSolder	~
	1	1 uF				Capacitor_SMD:C_0504_1310Metric_Pad0.83x1.28mm_HandSolder	~
	1	RED				Diode_SMD:D_0603_1608Metric_Pad1.05x0.95mm_HandSolder	~
	1	1N4148WT				Diode_SMD:D_SOD-523	https://ww
	1	2.2 kΩ				Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder	~
	1	470 Ω				Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder	~
	2	10 kΩ				Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder	~
	1	MCP73831-2-OT				Package_TO_SOT_SMD:SOT-23-5	http://ww1
	1	~				footprints:Arduino_Nicla_Sense	
	1	W25Q128JVS				Package_SO:SOIC-8_5.3x5.3mm_P1.27mm	https://ww

Next Week

Plans & Blockers for Next Week

- Test and finalized Rev. 3 PCB to Hand to Agriculture
- Add User Input/GUI to Testing Environment
- Obtain Powercast Transmitters from Agriculture Department
- Verify Circuit from Powercast

Thank You

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